

Evidence-Qualified Reliability–Energy Decision Boundaries for SRAM Protection

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Abstract—Selecting SRAM protection from area, timing, energy, and carbon estimates is unsafe when the estimates do not describe the same implementation or evidence boundary. We present a fail-closed decision method that admits an architecture only after its protection service, implementation identity, routed setup and hold feasibility, operation-resolved activity, and lifecycle scope are mutually consistent. We apply it to an unprotected baseline, two SECEDED organizations, and a stronger BCH identity in a pinned open SKY130 flow. The primary energy population reduces to 4 fully timing-clean matched SECEDED pairs and 40 post-route ECC-logic records. Hsiao is lower energy in only 4 of 20 paired operation cells; the sign depends on operation mix. The routed BCH identity provides stronger declared correction but is setup-clean in 0/5 seeds at the common target. We derive a workload boundary and a symbolic lifecycle break-even condition, while leaving SRAM-macro energy, physical failure rates, and SKY130 embodied carbon explicitly unresolved. Thus the result is a qualified decision region, not a global “greenest ECC” ranking.

Index Terms—SRAM, error-correcting codes, post-route activity, energy, carbon accounting, evidence qualification

I. INTRODUCTION

Memory-ECC comparisons commonly place correction capability, implementation cost, and energy on one Pareto plot. Yet a point may combine a logical guarantee from one code, timing from another wrapper, vectorless power, and a manufacturing coefficient from a different process boundary. The resulting precision is numerical rather than evidential. This problem becomes acute for sustainability: operational energy is workload-dependent, while embodied carbon depends on process, yield, geography, and system boundary [1], [2], [3], [4].

We ask: *which SRAM-protection preferences remain identifiable after categorical service, exact implementation identity, full timing feasibility, operation energy, and lifecycle assumptions are joined without imputing missing quantities?* Our contributions are: (i) a fail-closed evidence-admission procedure; (ii) a matched open-flow study that retains infeasible and incomplete candidates as negative results; and (iii) measured-coefficient workload and symbolic lifecycle decision boundaries. We do not claim a new code, activity model, global winner, field failure rate, or node-calibrated lifecycle result.

II. PRIOR WORK AND GAP

Hsiao established sparse odd-weight-column SECEDED construction [5]; Ghosh *et al.* showed that parity-check structure and input ordering affect memory-checker power under traces [6]. Cache-ECC and low-delay-code studies establish

reliability–area–performance trade-offs and hardware-aware construction [7], [8]. Adaptive protection is also prior art [9], as is activity-aware power estimation [10]. OpenROAD and OpenRAM provide inspectable implementation infrastructure [11], [12].

These works do not make it valid to merge unlike evidence. Physical upset topology is a required link between cell events and logical code service [13], [14]; semiconductor LCA likewise requires aligned scope and assumptions [15], [2], [3]. Our gap is therefore methodological and empirical: preserve the evidence boundary while determining which preferences are actually identifiable. A focused source map and rejected novelty claims are archived with the artifact.

III. EVIDENCE-QUALIFIED METHOD

A. Admission rule

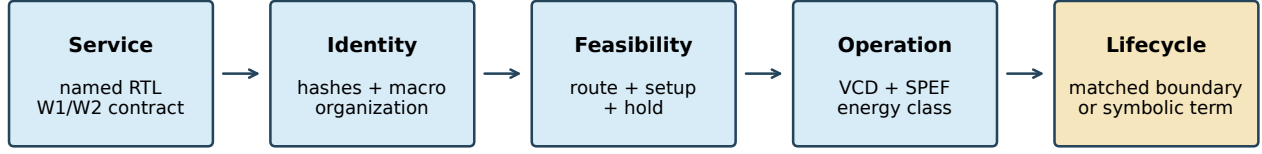
For architecture a , seed s , and operation o , define the admissibility indicator

$$A_{a,s,o} = G_a I_{a,s} T_{a,s} V_{a,s,o} B_{a,s,o}, \quad (1)$$

where G validates the declared protection service, I fixes RTL, macro organization, constraints, and hashes, T requires routed setup and hold slack to be nonnegative, V requires workload activity and extracted parasitics, and B requires a common measurement boundary. Every factor is Boolean; an unknown factor is zero rather than an invitation to substitute. Comparisons use only matched cells with $A = 1$ (Fig. 1).

B. Population and correctness boundary

Table I lists the implemented population. W1 and W2 denote weight-one and weight-two codeword errors; C and D denote correction and detection. The named extended-Hamming and Hsiao RTLs use a 64-bit payload and 72-bit codeword. Fresh regressions cover no-error and every W1/W2 mask over four fixed payloads, backed by frozen exact evidence. The shortened BCH(78,64, $t = 2$) corrects W1/W2 under its recorded boundary: all masks over four payloads plus 3082 fixed-mask arbitrary-symbolic-payload proofs. These are categorical services, not FIT or SDC-rate claims. A candidate SEC–DAEC RTL failed a fresh supported-mask test; a pipelined SECEDED lacked a validated memory controller; both were excluded before physical comparison.



Fail closed: infeasible and missing quantities remain visible; they do not enter the Pareto set.

Fig. 1. Evidence is admitted left-to-right. A missing or failed gate remains visible but cannot be repaired by a value from a different implementation or system boundary.

TABLE I
ADMITTED ARCHITECTURE POPULATION AT 10 NS

Identity	Service	Stored	Lat./II	Setup	E5
U0	none	64	1/1	5/5	not run
Ext.-Hamming	W1 C / W2 D	64+8	1/1	5/5	primary
Hsiao	W1 C / W2 D	64+8	1/1	5/5	primary
BCH(78,64,t=2)	W1-W2 C	64+16 ^a	1/1	0/5	not run

^aThe 78-bit codeword uses two 8-bit check macros and two padding bits. “Setup” counts clean seeds; E5 is post-route ECC logic, not macro-internal energy.

TABLE II
PRIMARY E5 ENERGY DELTA $E_H - E_C$ (PJ/OP)

Operation	Mean	Seed range	H lower
Idle	+0.0191	[-0.0287, +0.0681]	1/4
Clean read	-0.0812	[-0.2131, +0.1115]	3/4
Clean write	+0.1207	[+0.0730, +0.1395]	0/4
W1 correct	+0.2565	[+0.1483, +0.4770]	0/4
W2 detect	+0.1405	[+0.0298, +0.3554]	0/4

C. Physical and activity experiment

All retained physical records use the pinned ORFS/OpenROAD SKY130HD flow at the TT, 25°C, 1.80-V corner. Each architecture was routed at 5-ns and 10-ns targets with seeds 11, 13, 17, 19, and 23. Final reports provide placement, routing, parasitics, setup/hold timing, and route-DRC counts. Inherited SRAM22 256×64 and 256×8 LEF/GDS/Liberty views preserve macro identity; their independent DRC/LVS and complete internal energy are not established.

For conventional and Hsiao SECDED, operation traces contain 16 warm-up cycles followed by $N = 256$ measured operations. Zero-delay activity from the final routed netlist is joined to final SPEF. For the reported boundary,

$$E_{a,s,o} = P_{a,s,o} T_{s,o}/N, \quad \Delta E_o = \bar{E}_{H,o} - \bar{E}_{C,o}. \quad (2)$$

All 72 required SRAM-output roots are annotated, but E excludes SRAM-macro internal energy. The primary cohort is derived, not hand-selected: both implementations must be setup- and hold-clean and all five operation classes must exist.

IV. RESULTS

A. Feasibility precedes energy

All retained runs route to GDS with zero reported route-DRC errors. At 10 ns, U0 and conventional SECDED are setup/hold-clean in every seed. Hsiao is setup-clean in every seed but has a -0.0137-ns hold WNS in seed 11; the primary matched set is therefore seeds 13, 17, 19, 23. The unpipelined BCH identity is setup-clean in 0/5 seeds with mean WNS -11.112 ns (Fig. 2a). Stronger categorical service does not rescue an infeasible implementation.

Within the primary pair, Hsiao adds +222.4 μm^2 mean standard-cell area, +2868.0 μm routed wire, and +449.2 vias, while its mean setup WNS is +0.218 ns higher. These are logic/wrapper and macro-interface results; macro area is not attributed to the ECC logic.

B. Operation mix changes the preference

The primary cohort contains 40 E5 records and 20 paired cells. Hsiao is lower in 4 cells and conventional SECDED in 16. Clean read has a negative mean delta, whereas clean write, W1 correction, and W2 detection have positive deltas (Table II). A single scalar “ECC energy” therefore conceals a decision variable.

Let workload fractions p_I, p_R, p_W, p_C, p_D be nonnegative and sum to one. The measured primary-cohort boundary is

$$\Delta E(\mathbf{p}) = +0.0191p_I - 0.0812p_R + 0.1207p_W + 0.2565p_C + 0.1405p_D \text{ pJ/op}. \quad (3)$$

Hsiao is lower only when $\Delta E < 0$. For clean operations ($p_C = p_D = 0$), Fig. 3 shows the simplex partition derived from Eq. (3); no workload frequency is assumed.

C. What the carbon layer can and cannot say

For positive grid carbon intensity I_g , operational-carbon ordering has the same sign as Eq. (3). With an unknown embodied difference ΔC_{emb} and N_{life} lifetime operations,

$$\Delta C_{\text{life}} = \Delta C_{\text{emb}} + N_{\text{life}} I_g \Delta E(\mathbf{p}). \quad (4)$$

If $\Delta E < 0$ and $\Delta C_{\text{emb}} > 0$, the symbolic break-even is $N^* = \Delta C_{\text{emb}}/[-I_g \Delta E]$. The repository lacks a matched SKY130 manufacturing inventory, so no numeric ΔC_{emb} or lifecycle winner is reported. Published advanced-node models inform the equation and boundary audit only [1], [2], [4].

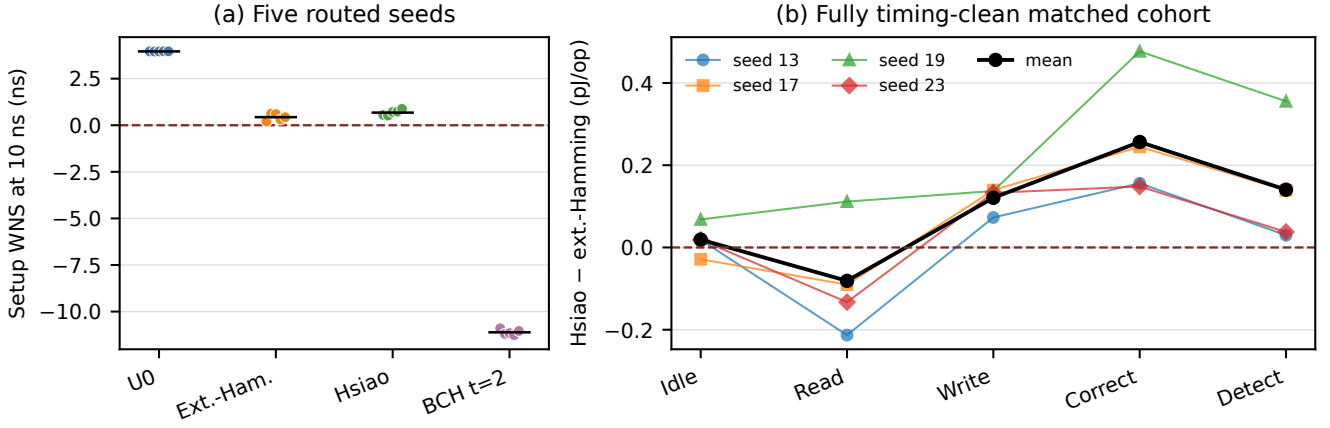


Fig. 2. (a) Setup WNS for the routed 10-ns population; bars mark seed means. (b) Hsiao-minus-extended-Hamming E5 ECC-logic energy in the four fully timing-clean matched pairs. Negative values favor Hsiao. Seed dispersion is descriptive, not a statistical population sample.

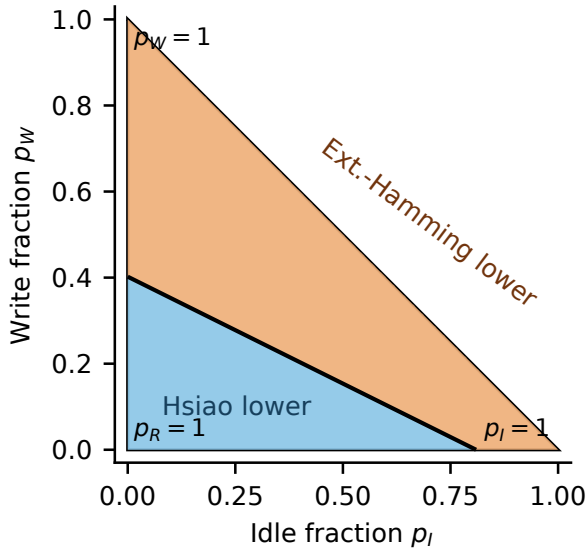


Fig. 3. Workload decision regions for idle, clean-read, and clean-write operations. The boundary uses source-derived four-seed mean coefficients.

V. DISCUSSION AND THREATS TO VALIDITY

A. Reading the decision surface

The method changes the interpretation of negative evidence. U0 cannot serve a W1-correction requirement; BCH is stronger but timing-infeasible as implemented; seed 11 is not silently retained after Hsiao's hold miss; and incomplete macro energy does not become whole-memory energy. The remaining comparison is narrow but internally consistent.

This ordering of gates is intentional. A reliability requirement first removes architectures whose declared service is insufficient. The physical gate then tests whether that service is delivered by the named implementation at the target constraint. Only equal-boundary, timing-clean implementations enter the operation-energy comparison. Finally, carbon accounting dis-

tinguishes three outcomes: an operational preference when $\Delta E(\mathbf{p})$ has known sign, a symbolic lifecycle crossover when embodied difference is unknown, and an indeterminate result when measurement boundaries differ. Reporting these outcomes separately prevents a low-cost unprotected design or a stronger but unusable decoder from appearing on a misleading common front.

The clean-read region in Fig. 3 is not a recommendation to select Hsiao unconditionally. It says exactly what additional evidence a design decision needs: a credible operation mix, the intended clock constraint, and a macro-energy characterization at the same activity boundary. Conversely, the positive correction/detection coefficients matter even when faults are rare, because scrub, test, or recovery policies can change their frequency. The equation exposes that policy coupling without pretending to know it.

The BCH result should also be read as implementation-specific. The correctness evidence admits its stronger categorical service, while physical evidence rejects this unpipelined identity at the common target. Pipelining or iterative decoding could change feasibility and energy, but also changes latency, control, and RTL identity. It therefore requires a new transaction wrapper and a new matched experiment; it cannot inherit the present record.

B. Reproducibility contract

The artifact separates immutable campaign evidence from paper derivations. A single script hashes and reads the physical CSV, architecture audit, activity matrix, campaign status, and run manifest; derives the timing-clean cohort; recomputes all energy and physical deltas; and emits the figures, tables, manuscript macros, and claim-level provenance CSV. The full campaign contains 46 activity-qualified records, but the paper's primary numbers are generated only from the 40 complete records. No RTL-to-GDS rerun is required to reproduce the paper because the paper reports the frozen campaign rather than a newly sampled implementation population. Exact commands,

source hashes, units, formulas, and the distinction between reproduction and expensive campaign regeneration accompany the manuscript.

C. Threats to validity

The experiment covers one PDK/library/corner, two clock targets, five deterministic route seeds, one RTL identity per architecture, and synthetic operation classes. Seeds describe implementation sensitivity, not a random sample, so no confidence intervals or significance claims are made. Zero-delay VCD omits delay-induced glitch activity. SRAM macro-internal energy, independently validated macro signoff, Qcrit, physical bit mapping, radiation environment, FIT/SDC/DUE, and lifecycle inventory remain open. A new macro characterization, pipelined BCH wrapper, or deployed workload could move the boundary; each would be a new evidence record, not a retroactive correction.

VI. CONCLUSION

Evidence qualification turns an apparent four-way sustainability ranking into a sequence of conditional decisions. The stronger BCH identity fails the common timing gate, and the equal-service SECDED comparison reduces to 4 full-timing-clean pairs. Across their 20 operation cells, Hsiao is lower energy in only 4, with the workload boundary given explicitly rather than collapsed into a winner. Absolute lifecycle ordering remains symbolic because macro energy, physical reliability rates, and SKY130 embodied carbon are not qualified. The practical result is not less information: it is a decision surface whose valid and invalid regions are auditable.

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